Kusuma Sai

Software Engineer – Full Stack & AI
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Education

National Institute of Technology, Rourkela *2019 – 2023*
B.Tech, Computer Science and Engineering – CGPA: 8.32/10.00
Coursework: DSA, DBMS, Operating Systems, AI & ML, Software Engineering, OOP,Computer Networks

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL,NOSQL
Full-Stack & Backend: Spring Boot,Hibernate, React.js, Node.js, Tomcat
Data & Messaging: MongoDB, Postgres, Kafka
AI / LLM Engineering: RAG, Agentic AI, MCP Servers, Prompt Engineering
Test Automation: Playwright, JUnit
DevOps & Tooling: Git, Gradle, Maven, Harness, Jira, OCP, CI/CD pipelines, kubernetes
Certification: Microsoft Azure AI Fundamentals (AI-900)

Experience

Wells Fargo – Software Engineer *Jul 2024 – Present*
Case Investigation Tool-enhancements

- Independently diagnosed and refactored tightly coupled data structures causing 504 gateway timeouts, **cutting latency 6X and eliminating 2-5 production incidents/month**.
- Implemented app-to-app and user-to-app authentication/authorization using **Ping** and **APIGEE**.
- Integrated AI tools **GitHub Copilot, Claude, and MCP servers** into daily workflow, streamlining ticket triage, PR reviews,remediating vulnerabilities,upgrades and testing.

Case Investigation Tool-sunrise

- Designed service boundaries and API contracts for a **5-microservice system (Java, Spring Boot, Kafka, IBM BAM, React)**, owning **40% of the architecture decisions**; delivered structured investigation workflows and full audit trails used by 50 agents/teams,with 100 cases per day.
- Owned a **resilient data ingestion pipeline** end-to-end, implementing **circuit breaker, retry, and idempotency** patterns.Currently **processing 50k events per hour**.
- used **Kafka** for high-volume real-time case ingestion; used **MongoDB** to handle dynamic, unstructured case data.
- Partnered with business stakeholders, solution architects, and control partners from requirements through production delivery.

Wells Fargo – Program Associate *Jul 2023 – Jul 2024*
Application Modernization

- Co-led a Jakarta EE migration from **Java 8/Spring 5/JBoss to Java 17/Spring Boot 3/Tomcat 10,** reverse-engineering undocumented legacy code to isolate hidden dependencies and prevent regressions for an *internal distributed system with 300 users.*
- Diagnosed and resolved version-incompatibility issues (deprecated APIs, classloader conflicts, Hibernate/Spring Boot defaults) mid-migration, validating each fix against existing behavior.
- **Eliminated $150K** in annual DB2 licensing costs and removed risk from IBM support withdrawal by rewriting core web app, batch, and Quartz jobs.

InternshipStudio – Machine Learning Intern *Jun 2022*

- Built a machine learning model to predict YouTube ad views, focused on accuracy and efficiency.
- Performed data preprocessing, feature engineering, and model tuning with Pandas, NumPy, and Scikit-learn.

AI/ML Projects

Content Automation Pipeline – AI Workflow *Jun 2026*

- Built an end-to-end pipeline using **Gemini 2.5 Flash** with prompt-engineered structured JSON output to auto-generate and publish daily AI-education carousel content.
- Built a custom Python Pillow image engine rendering branded 1080x1350 slides, deployed via Cloudinary and the Instagram Graph API.

Autonomous Test Data & Execution Agent – Playwright + Custom MCP *Jan 2026*

- Built an end-to-end autonomous agent that resolves all test-data variables against a spec before Playwright execution, using a custom MongoDB MCP server to query or generate records with enforced per-variable uniqueness to prevent cross-test collisions.
- Randomized data generation uncovered latent functional bugs masked by static test data, improving defect detection beyond the original suite.

Financial Rules Analyzer – Hackathon *Mar 2025*

- Automated data profiling and adaptive risk scoring using **DeepSeek** to streamline compliance review and cut manual effort.
- Enabled AI-driven rule generation from regulatory guidelines with context-aware remediation suggestions, built on Streamlit, Python, and the DeepSeek API.

Achievements

- Won **Spotlight Award 3x** at Wells Fargo for Execution Excellence,and proactively taking preventive measures against production incidents and providing quick resolution avoiding business impact.